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Projects on Deep Learning for Phase Extraction in Digital Holography

Somesh Nandi <someshnandi@rvce.edu.in> Sat, Jul 19, 2025 at 9:16 PM
To: Artificial Intelligence and Machine Learning Students Group 2023 <ai_group_2023@rvce.edu.in>

Dear Students,

There are some **advanced projects in the domain of Digital Holography and Deep Learning, are available to explore in the department**

The following **potential tasks** are available:

1. **Test dataset generation** (wrapped phase) using mathematical equations of interferograms corrupted by noise (AWGN, speckle)
2. **De-noising of noisy wrapped phase** using generative or transformer-based deep learning models
3. **One-step phase extraction** from noisy interferograms using generative or transformer-based models
4. **Simultaneous extraction of phase and phase derivatives** using **diffusion models**

These projects involve **a significant amount of mathematics**, particularly in areas such as **linear algebra, calculus, Fourier transforms, and signal processing.**

Familiarity with Python and deep learning frameworks like TensorFlow or PyTorch is desirable.

If you are not comfortable with mathematical modeling and theory, I kindly request you to avoid applying for these topics and ignore the email, as a strong mathematical foundation is a prerequisite.

If you're genuinely interested, please feel free to reach out to me.

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